

TripStack API Automation – Document

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Project

TripStack API Automation Framework

Technology Stack

- Java
 - Rest Assured
 - JUnit 5
 - Gradle
 - IntelliJ IDEA
-

Objective

Automate the assigned booking journey provided in the capstone exam.

Employee Details

- Employee ID : **1015**
- User : **Olivia Om**
- Email : **olivia@tripstack.test**

Assigned Journey

Flight Multi-City

- COK → BOM
 - BOM → DEL
-

Framework Structure

src

└─ test

└─ java

└─ com.apitestng

```
└─ client
└─ config
└─ data
|   └─ builder
|   └─ model
|   └─ secrets
└─ support
└─ tests
```

Implemented Features

1. Login

Automated login using Rest Assured.

Request

POST /api/auth/login

Assertions

- Status Code = 200
 - JWT Token generated
 - Token stored for remaining API calls
-

2. Booking Creation

Created bookings for the assigned multi-city journey.

First Leg

COK → BOM

Inventory

FL-COKBOM-51

Second Leg

BOM → DEL

Inventory

FL-BOMDEL-52

Assertions

- Status Code = 201
- Booking State = HELD

3. Dynamic Seat Selection

Implemented random seat selection.

Features

- Random seat generation
- Avoid duplicate seats in same execution
- Retry automatically if

409 SEAT_CONFLICT

is returned.

The framework keeps selecting another seat until booking succeeds.

4. Payment

Automated payment.

Request

POST

/api/bookings/{bookingId}/pay

Assertions

- Status Code = 200
- State becomes

PAYMENT_PENDING

5. Booking Confirmation

Automated confirmation.

Request

POST

/api/bookings/{bookingId}/confirm

Assertions

- Status Code = 200
- State becomes

CONFIRMED

PNR is extracted and stored.

6. Booking Retrieval

Verified booking after confirmation.

Request

GET

/api/bookings/{pnr}

Assertions

- Status Code = 200
 - Correct PNR
 - State = CONFIRMED
-

7. Namespace Validation

Verified that generated PNR belongs to Employee 1015.

Expected format

TS-1015-XXXX

Assertions

- PNR starts with

TS-1015-

This confirms the booking belongs to the logged-in employee namespace.

Employee Namespace

The framework logs in using

olivia@tripstack.test

The JWT contains

empld = 1015

Every booking created belongs only to Employee 1015.

PNRs are validated against this namespace.

Error Handling

Implemented automatic retry for seat conflicts.

Scenario

409 SEAT_CONFLICT

Framework behavior

- Detect conflict
- Pick another random seat
- Retry booking automatically
- Continue until booking succeeds

No manual intervention is required.

API Workflow

Login

↓

Create Booking (Leg 1)

↓

Create Booking (Leg 2)

↓

Payment (Leg 1)

↓

Payment (Leg 2)

↓

Confirm (Leg 1)

↓

Confirm (Leg 2)

↓

Get Booking (Leg 1)

↓

Get Booking (Leg 2)

↓

Validate Employee Namespace

Assertions Implemented

Login

- Status Code
- Token generated

Create Booking

- Status Code = 201
- State = HELD

Payment

- Status Code = 200
- State = PAYMENT_PENDING

Confirmation

- Status Code = 200
- State = CONFIRMED

Get Booking

- Status Code = 200
- Correct PNR
- State = CONFIRMED

Namespace

- PNR starts with

TS-1015-

Current Status

Module	Status
Login	Completed
Booking Creation	Completed
Multi-City Journey	Completed
Dynamic Seat Selection	Completed
Seat Conflict Retry	Completed
Payment	Completed
Confirmation	Completed
Booking Retrieval	Completed
Namespace Validation	Completed

Summary

The API automation framework successfully automates the assigned **Flight Multi-City** booking journey for **Employee ID 1015 (Olivia Om)**. It authenticates the user, creates bookings for both flight legs, handles seat conflicts automatically by selecting new seats, completes payment and confirmation, retrieves booking details, and validates that the generated PNR belongs to the correct employee namespace